

Problem: Determine a one-parameter Ritz solution for an isotropic rectangular plate with edges $x = 0, a$ clamped and edges $y = 0, b$ simply supported, and subjected to a transverse center point load P_0 . Use the approximated solution in Eq. (1). Determine the maximum deflection of the rectangular plate. The thickness, modulus of elasticity and Poisson ratio of the plate are h , E and ν , respectively.

$$w(x, y) = c \left(1 - \cos \left(\frac{2\pi x}{a} \right) \right) \sin \left(\frac{\pi y}{b} \right) \quad (1)$$

Solution

A one-parameter function is given as

$$w(x, y) = c \left(1 - \cos \frac{2\pi x}{a} \right) \sin \frac{\pi y}{b}$$

First, we should calculate derivatives of the function to check essential boundary conditons.

$$\begin{aligned} \frac{\partial w}{\partial x} &= c \left(\frac{2\pi}{a} \sin \frac{2\pi x}{a} \right) \sin \frac{\pi y}{b} \\ \frac{\partial w}{\partial y} &= c \left(1 - \cos \frac{2\pi x}{a} \right) \left(\frac{\pi}{b} \cos \frac{\pi y}{b} \right) \\ \frac{\partial^2 w}{\partial x^2} &= c \left(-\frac{4\pi^2}{a^2} \cos \frac{2\pi x}{a} \right) \sin \frac{\pi y}{b} \\ \frac{\partial^2 w}{\partial y^2} &= c \left(1 - \cos \frac{2\pi x}{a} \right) \left(-\frac{\pi^2}{b^2} \sin \frac{\pi y}{b} \right) \\ \frac{\partial^2 w}{\partial x \partial y} &= c \left(\frac{2\pi}{a} \sin \frac{2\pi x}{a} \right) \left(\frac{\pi}{b} \cos \frac{\pi y}{b} \right) \end{aligned}$$

So, we have clamped and simply supported boundary conditions;

- Clamped at $x = 0, a$: $w = 0$ and $\frac{\partial w}{\partial x} = 0$,

$$\begin{aligned} \text{At } x = 0: \quad w &= c \left(1 - \cos 0 \right) \sin \frac{\pi y}{b} = 0 \\ \frac{\partial w}{\partial x} &= c \left(\frac{2\pi}{a} \sin 0 \right) \sin \frac{\pi y}{b} = 0 \\ \text{At } x = a: \quad w &= c \left(1 - \cos 2\pi \right) \sin \frac{\pi y}{b} = 0 \\ \frac{\partial w}{\partial x} &= c \left(\frac{2\pi}{a} \sin 2\pi \right) \sin \frac{\pi y}{b} = 0 \end{aligned}$$

- Simply supported at $y = 0, b$: $w = 0$,

$$\begin{aligned} \text{At } y = 0: \quad w &= c \left(1 - \cos \frac{2\pi x}{a} \right) \sin 0 = 0 \\ \text{At } y = b: \quad w &= c \left(1 - \cos \frac{2\pi x}{a} \right) \sin \pi = 0 \end{aligned}$$

All essential conditions are satisfied. Now we want to find the total potential energy;

$$\Pi = \frac{D}{2} \int_0^a \int_0^b \left[(w_{xx} + w_{yy})^2 - 2(1 - \nu)(w_{xx}w_{yy} - w_{xy}^2) \right] dx dy - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \quad (2)$$

Substituting the obtained derivatives into Eq. (2) gives

$$\begin{aligned} & \frac{D}{2} \int_0^a \int_0^b \left[(w_{xx} + w_{yy})^2 - 2(1 - \nu)(w_{xx}w_{yy} - w_{xy}^2) \right] dx dy - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \\ &= \frac{D}{2} \int_0^a \int_0^b \left[\left(\frac{4P_0}{\pi^2 D} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right)^2 - 2(1 - \nu) \left(\frac{4P_0}{\pi^2 D} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right) \right. \\ & \quad \left. - 2(1 - \nu) \left(\frac{4P_0}{\pi^2 D} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right) \right] dx dy - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \\ &= \frac{D}{2} \int_0^a \int_0^b \left[\frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right)}{(m^2 + n^2)^2} - 2(1 - \nu) \frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right] dx dy \\ & \quad - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \\ &= \frac{D}{2} \int_0^a \int_0^b \left[\frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right)}{(m^2 + n^2)^2} - 2(1 - \nu) \frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right] dx dy \\ & \quad - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \\ &= \frac{D}{2} \int_0^a \int_0^b \left[\frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right)}{(m^2 + n^2)^2} - 2(1 - \nu) \frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right] dx dy \\ & \quad - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \\ &= \frac{D}{2} \int_0^a \int_0^b \left[\frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin^2\left(\frac{m\pi x}{a}\right) \sin^2\left(\frac{n\pi y}{b}\right)}{(m^2 + n^2)^2} - 2(1 - \nu) \frac{16P_0^2}{\pi^4 D^2} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)}{m^2 + n^2} \right] dx dy \\ & \quad - P_0 w\left(\frac{a}{2}, \frac{b}{2}\right) \end{aligned}$$

The work done by the point load is

$$W = P_0 w\left(\frac{a}{2}, \frac{b}{2}\right)$$

Integrating the expressions, we obtain

$$\begin{aligned} & \int_0^a \int_0^b \left(\frac{2b}{a^3} + \frac{1}{ab} + \frac{3a}{8b^3} \right) dx dy \\ &= \int_0^a \left[\left(\frac{2b}{a^3} + \frac{1}{ab} + \frac{3a}{8b^3} \right) y \right]_0^b dy \\ &= \int_0^a \left(\frac{2b}{a^3} + \frac{1}{ab} + \frac{3a}{8b^3} \right) b dy \\ &= \int_0^a \left(\frac{2b^2}{a^3} + \frac{1}{a} + \frac{3ab}{8b^3} \right) dy \\ &= \int_0^a \left(\frac{2b^2}{a^3} + \frac{1}{a} + \frac{3a}{8b^2} \right) dy \\ &= \left[\left(\frac{2b^2}{a^3} + \frac{1}{a} + \frac{3a}{8b^2} \right) y \right]_0^b \\ &= \left(\frac{2b^2}{a^3} + \frac{1}{a} + \frac{3a}{8b^2} \right) b \end{aligned}$$

Simplifying the total potential energy,

$$\begin{aligned} & U = \frac{1}{2} P_0 c \int_0^a \int_0^b \left(\frac{2b}{a^3} + \frac{1}{ab} + \frac{3a}{8b^3} \right) dx dy \\ &= \frac{1}{2} P_0 c \left(\frac{2b^2}{a^3} + \frac{1}{a} + \frac{3a}{8b^2} \right) b \end{aligned}$$

$$w_{max} = \frac{24P_0(1-\nu^2)}{\pi^4 E h^3 \left(\frac{2b}{a^3} + \frac{1}{ab} + \frac{3a}{8b^3} \right)}$$